Tab 1

Five Blogs

1. **Increase access to resilient and affordable urban services and infrastructure**

Systems Change Lab. (n.d.). *Increase access to resilient and affordable urban services and infrastructure*. https://systemschangelab.org/cities/improve-urban-infrastructure-services

Key points:

- About **1 in 3 urban residents lack access to basic services** like water, sanitation, energy, and transport.
- The "urban services divide" disproportionately affects poorer communities.
- Urban growth and climate risks are intensifying pressure on service systems.

2. **The Utilities of Cities**

World Bank. (n.d.). *The utilities of cities*. World Bank Blogs. https://blogs.worldbank.org/en/sustainablecities/the-utilities-of-cities

Key points:

- Urban service delivery (utilities, transport, energy) is becoming **more complex and data-driven**.
- Cities face rising risks from **climate shocks and infrastructure failures**.
- Utilities must evolve to handle **growing demand and shrinking budgets**

3. **Transformations to Solve Urban Inequality**

World Resources Institute. (n.d.). *Transformations to solve urban inequality*. https://www.wri.org/insights/transformations-to-solve-urban-inequality

Key points:

- Many city residents rely on **informal or costly alternatives** due to lack of public services.
- Poor service provision leads to **higher costs, worse health, and environmental damage**.
- Calls for **integrated, cross-sector urban service reforms**.

4. **Reliable and inclusive urban sanitation: A driver of growth** World Bank. (2025). *Reliable and inclusive urban sanitation: A driver of growth*. World Bank Blogs. https://blogs.worldbank.org/en/water/reliable-and-inclusive-urban-sanitation--a-driver-of-growth

Key points:

- **3.5 billion people lack safe sanitation**, a critical urban service.
- Strong sanitation systems improve **health, education, and economic productivity**.
- Urban futures depend heavily on **basic service availability and management**.

5. **Urban governance is vital for cities to prosper**

United Nations Development Programme. (n.d.). *Urban governance is vital for cities to prosper*. https://www.undp.org/blog/urban-governance-vital-cities-prosper

Key points:

- **Governance is central to service delivery:** Effective urban governance determines how **services, infrastructure, and resources** are planned and distributed.
- **Rapid urbanization strains services:** Growing city populations increase pressure on **housing, transport, sanitation, and waste management systems**.
- **Service gaps are widespread:** Many cities face **inadequate infrastructure, informal settlements, and unequal access** to basic services

Additional Blogs

Weston, M., & King, R. (2021, October 19). *7 major transformations to solve urban inequality*. TheCityFix. Major Transformations to Solve Urban Inequality | TheCityFix

- The article explains the **"urban services divide,"** where many urban residents still lack access to basic services such as water, sanitation, housing, transportation, and electricity. This inequality worsens poverty and reduces quality of life.
- It emphasizes that cities should prioritize **inclusive and climate-resilient infrastructure**, especially for informal settlements and underserved communities that are often excluded from urban planning and public services.

- The article highlights the importance of **better urban planning, local data collection, and collaboration** between governments, communities, and alternative service providers to improve service delivery and reduce inequality.
- It concludes that equitable access to urban services can create **more sustainable, resilient, and economically productive cities**, benefiting both disadvantaged populations and society as a whole.

MacLeod, A. (2022, September 22). *"0 to 1 in sixty seconds" – Data's journey in shaping digital transformation in utilities*. Ground Control. https://www.groundcontrol.com/blog/datas-journey-in-shaping-digital-transformation-in-utilities/

- The article explains how digital transformation is improving utility services through data collection, IoT, and smart technologies.
- Smart meters and sensors help utility providers monitor energy usage, reduce outages, and improve efficiency.

Philippine Information Agency. (2026, May 5). DOST, UP launch SMART METRO to advance government services nationwide. https://pia.gov.ph/press-release/dost-up-launch-smart-metro-to-advance-government-services-nationwide/

- SMART METRO is a new digital system launched in Iloilo City (April 2026) consisting of four platforms—Connect (citizen reporting), Command (LGU response tracking), Collab (internal coordination), and Codex (data/mapping for planning)
- The system is designed to tackle disasters, emergency responses, traffic and mobility, land use and flooding, health concerns, and economic planning

Smart Growth America. (n.d.). The smart growth approach: More homes in the right places, at the right price. Smart Growth America Knowledge Hub. https://www.smartgrowthamerica.org/knowledge-hub/news/the-smart-growth-approach-more-homes-in-the-right-places-at-the-right-price/

- The US is short 4.7 million homes. The blog argues this crisis is "no accident" but rather the result of rules, policies, and choices that have sprawled housing outward, made it astronomically expensive, and offered little choice in the types of places people can call home.
- Emphasizes that transportation, land use, and economic development don't happen on their own. A bad decision in one area ripples into others, creating problems that last decades. Smart growth brings it all together so people can live in their communities at any stage of life.
- Reforming outdated zoning codes, rethinking land availability (reusing parking lots, vacant parcels, brownfields instead of building on green space), and identifying new financing tools to make more types of housing feasible in well-connected locations.

Aimsun. (n.d.). *Machine learning for traffic forecasting*. Aimsun. Machine Learning for Traffic Forecasting – Aimsun

Ratcliffe, R. (2022, December 13). *As waste-to-energy incinerators spread in Southeast Asia, so do concerns*. Mongabay. https://news.mongabay.com/2022/12/as-waste-to-energy-incinerators-spread-in-southeast-asia-so-do-concerns/

Five academic/journal papers

1. Gozun, B. C., & De Castro, J. M. (2026). Household perspectives on urban water service delivery: Implications for policy and practice in the Philippines. *Public Works Management & Policy*. Advance online publication. https://doi.org/10.1177/1087724X261439281

2. Bibri, S. E., & Krogstie, J. (2024). Exploring the symbiotic relationship between digital transformation, infrastructure, service delivery, and governance for smart sustainable cities. *Smart Cities, 7*(2), 34. https://doi.org/10.3390/smartcities7020034

3. Gerlach, E., Franceys, R., & Miles, J. (2023). The enabling environment for citywide water service provision: Insights from six successful cities. *PLOS Water, 2*(3), e0000071. https://doi.org/10.1371/journal.pwat.0000071

4. Moretto, L., Faldi, G., Ranzato, M., & Rosati, F. (2023). Coproduced urban water services: When technical and governance hybridisation go hand in hand. *Frontiers in Sustainable Cities, 4*, 969755. https://doi.org/10.3389/frsc.2022.969755

5. La Vigna, F. (2022). Urban groundwater issues and resource management, and their roles in the resilience of cities. *Hydrogeology Journal*, *30*(6), 1657-1683. https://doi.org/10.1007/s10040-022-02517-1

Additional Papers

Urban Traffic Congestion

- Rahman, M. M., Ahmed, S., & Kim, J. (2024). Urban traffic congestion prediction using machine learning techniques. *Smart Cities, 7*(1), 10. https://doi.org/10.3390/smartcities7010010

Urban Flooding

- Santos, R., Delgado, P., & Cruz, M. (2024). Climate risk and flood vulnerability in informal urban settlements. *Land, 13*(9), 1357. https://doi.org/10.3390/land13091357

Informal Settlement Housing

- Turok, I., & Borel-Saladin, J. (2023). Housing adequacy in informal built environments: Challenges and opportunities. *Journal of Housing and the Built Environment, 38*(2), 100–118. https://doi.org/10.1007/s10901-023-10029-x

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Public Transportation System

- Almeida, A., Brás, S., Sargento, S., & Oliveira, I. (2023). *Exploring bus tracking data to characterize urban traffic congestion*. **Journal of Urban Mobility, 4**, Article 100065. https://doi.org/10.1016/j.urbmob.2023.100065

Waste Management

Imran, M., Jijian, Z., Sharif, A., & Magazzino, C. (2024). Evolving waste management: The impact of environmental technology, taxes, and carbon emissions on incineration in EU countries. *Journal of Environmental Management*, *364*, 121440. **https://doi.org/10.1016/j.jenvman.2024.121440**

Bike Lanes

Li, D., & Lasenby, J. (2023). *Mitigating urban motorway congestion and emissions via active traffic management*. Research in Transportation Business and Management, 48, Article 100789. https://doi.org/10.1016/j.rtbm.2022.100789

Mixed-Use Urban Development

- Wei, Y. D., & Xiao, W. (2023). Urban density, land use, and socioeconomic characteristics: Implications for sustainable development. *Environment and Urbanization, 35*(2), 456–472. https://doi.org/10.1177/09562478231195512

Water Distribution Systems

- Williams, G., Thomas, E., & Singh, R. (2024). Finding space for water: Infrastructure challenges in informal settlements. *Environment and Urbanization ASIA, 15*(1), 1–15. https://doi.org/10.1177/24557471241282209

Slum Sanitation Conditions

- Zhang, Q., Li, H., & Chen, X. (2023). Environmental determinants of sanitation access in informal settlements. *Infectious Diseases of Poverty, 12*(1), 78. https://doi.org/10.1186/s40249-023-01078-z

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Smart City Dashboard / Systems

- Kumar, A., Singh, P., & Sharma, R. (2024). Smart parking systems and IoT-based urban mobility solutions. *Smart Cities, 7*(3), 52. https://doi.org/10.3390/smartcities7030052

Equality of access to urban facilities

Fan, C., Jiang, X., Lee, R., & Mostafavi, A. (2022). Equality of access and resilience in urban population-facility networks. *npj Urban Sustainability, 2*(1), 9. https://doi.org/10.1038/s42949-022-00051-3

Digital Transformation of Utility Services

Khot, S., Waghre, A. R., Abbas, M. H., & Zubair, M. (2024). *Digital transformation of utility services using AI, IoT and ITSM practices: Evidence from water, electricity and waste management systems*. *Journal of Information Systems Engineering and Management, 9*(4s). https://doi.org/10.13140/RG.2.2.14844.60801

Traffic Safety

Khan, M. N., & Das, S. (2024). Advancing traffic safety through the safe system approach: A systematic review. *Accident Analysis & Prevention*, *199*, 107518. https://www.sciencedirect.com/science/article/pii/S0001457524000630

Five videos

1. https://ocw.tudelft.nl/course-lectures/introduction-urban-services-infrastructures/
2. https://unhabitat.org/knowledge/global-urban-lectures
3. https://youtu.be/toEQZapHYLA?si=uJWbk0D41pOlzUqH
4. https://youtu.be/2kIS71JA1Lk?si=BqsjBsqEKbOMeZm6
5. https://youtu.be/Q1NLATszskU?si=4Y8VrpPAFAIx7op_
6. https://www.wri.org/insights/transformations-to-solve-urban-inequality
7. https://ocw.tudelft.nl/course-lectures/changing-urban-services-infrastructures/
8. Video Shorts: Urbanization – The Human Imprint
9. https://archive.metabolismofcities.org/videos/43-what-is-urban-metabolism
10. https://archive.metabolismofcities.org/videos/45-participatory-urban-metabolism-in-medellin-colombia
11. https://smartcity.go.kr/2017/10/03/smart-cities-building-for-the-cities-of-tomorrow-documentary-2015/
12. https://www.worldbank.org/en/news/video/2017/04/19/cities-are-where-the-future-is-being-built
13. https://unhabitat.lk/multimedia/state-of-sri-lankan-cities-report-multimedia/video-urban-sprawl-and-compact-cities/
14. https://youtu.be/0SoKj5Gxnnk?si=8FyWVB96UjgKFRgY
15. https://youtu.be/tgYOOPMQmc4?si=dA2Cqe3fs_ErdGM0
16. https://youtu.be/ni3teWmnOR0?si=qtt_XJnh5PxovGrg

Photos

ECN100 Research Folder

content - piaurrr

HOMEPAGE: The Urban Services Divide

Who Cities Work For—and Who They Don't

Project Overview

Urbanization is one of the most powerful economic drivers of the 21st century, yet its benefits are not distributed equally. This project explores the **Urban Services Divide**: the widening gap between those who have access to resilient, affordable infrastructure and those forced to rely on informal, high-cost alternatives. By analyzing the intersection of governance, economics, and climate resilience, we examine how urban services—from water and sanitation to transport and energy—dictate the economic mobility of billions.

BLOG SECTION: Critical Perspectives

1. Bridging the Gap: Resilient Infrastructure for All

About 1 in 3 urban residents globally lack access to essential services. This is not merely a logistical failure; it is a systemic economic barrier.

- **The Economic Mechanism:** When infrastructure is not resilient, a single climate event (like a flood) can decapitalize a poor household by destroying their only assets.
- **The Divide:** Wealthier districts benefit from centralized, subsidized grids, while informal settlements pay a "poverty premium"—paying more per unit for water or electricity from private vendors than those with formal connections.
- **Strategic Focus:** Transitioning to "Systems Change" requires moving beyond piecemeal projects to integrated urban planning that treats infrastructure as a human right and an economic necessity.

2. The Digital and Data Evolution of City Utilities

Modern utilities are no longer just about pipes and wires; they are about data flows.

- **The Complexity Shift:** As cities face shrinking budgets, "Smart Utility" management (using IoT and AI) becomes essential to reduce "non-revenue water" (leaks) and optimize energy grids.
- **The Risk Factor:** Without data-driven management, cities are "blind" to infrastructure failures until they become catastrophes. However, a digital divide persists: if data only tracks formal residents, the informal population remains invisible to the planners.

3. Structural Transformations to Solve Urban Inequality

Inequality is "baked in" to the physical layout of many cities.

- **The Hidden Costs:** Lack of public transport doesn't just mean a long commute; it means restricted access to the labor market.
- **Cross-Sector Reform:** We cannot solve housing without solving sanitation, and we cannot solve sanitation without solving water. An integrated approach recognizes that improvements in one sector (e.g., better drainage) drastically reduce costs in others (e.g., public health expenditures).

4. Sanitation: The Invisible Engine of Growth

With 3.5 billion people lacking safe sanitation, this remains the most neglected urban service.

- **Productivity Impacts:** Inadequate sanitation leads to stunting and chronic illness, which permanently reduces the lifetime earning potential of the urban poor.
- **The Future of Cities:** Safe fecal sludge management (FSM) is a driver of the "Circular Economy," where waste can be converted into energy or fertilizer, creating new urban value chains.

5. Governance: The Blueprint for Prosperity

Infrastructure is only as good as the institutions that manage it.

- **Planning vs. Reaction:** Rapid urbanization often outpaces planning, leading to "reactive governance."
- **The Role of UNDP:** Effective governance ensures that resource distribution is transparent. When governance fails, the vacuum is filled by informal cartels that exploit the vulnerable, further widening the service divide.

ACADEMIC COMPENDIUM: Synthesis of Research

Core Water & Governance Papers

- **Gozun & De Castro (2026):** Offers a "bottom-up" view of the Philippines, proving that household satisfaction is tied not just to the *presence* of water, but to its *reliability*. Policy must shift from "access" to "quality of service."
- **Bibri & Krogstie (2024):** Explores the "Symbiotic Relationship" of the Smart City. They argue that digital transformation is the "glue" that connects governance to service delivery, making sustainable cities possible.
- **Gerlach et al. (2023):** Identifies the "Enabling Environment." Success in water provision isn't just about engineering; it's about the legal and financial frameworks that allow utilities to function autonomously and efficiently.

Thematic Research: Challenges & Solutions

Theme	Key Finding	Economic/Social Impact
Traffic (Rahman et al.)	Machine learning can predict congestion.	Reduces lost "man-hours" and lowers urban carbon footprints.
Flooding (Santos et al.)	Informal settlements are disproportionately in flood zones.	Climate change acts as an "inequality multiplier."
Housing (Turok & Borel-Saladin)	Adequacy is more than four walls.	Proper housing is the "anchor" for all other urban services.
Waste (Imran et al.)	Environmental taxes drive incineration tech.	Transitions waste from a liability to a resource (energy).

MULTIMEDIA GALLERY: Visualizing the Imprint

Lecture Series & Documentaries

- **Urban Metabolism (Metabolisms of Cities):** These videos explain cities as living organisms. Understanding the "input" (resources) and "output" (waste) is key to designing circular, resilient services.
- **World Bank: Cities of the Future:** Highlights that by 2050, 70% of the population will live in cities. The infrastructure we build *today* determines the climate outcomes of *tomorrow*.
- **UN-Habitat Lectures:** A deep dive into the "Compact City" model, which argues that higher density (if managed well) makes service delivery more affordable and efficient.

SYNTHESIS FOR RESEARCH FOLDER (ECN100)

Key Takeaway for Policy Makers: To close the urban services divide, interventions must be:

1. **Inclusive:** Targeting informal settlements as part of the formal city fabric.
2. **Data-Driven:** Using smart systems to manage scarce resources.
3. **Cross-Sectoral:** Recognizing that water, energy, and transport are interdependent systems.

Source Links Included in Project:

- World Bank Sustainable Cities
- WRI: Transformations to Solve Inequality
- Smart Cities Journal (Bibri & Krogstie)
- Video: Urban Metabolism

fleshed out content - piaurrr

Who Cities Work For—and Who They Don't: The Urban Services Divide

Project Overview

Urbanization is one of the most powerful economic drivers of the 21st century, yet its benefits are not distributed equally. This project explores the **Urban Services Divide**: the widening gap between those who have access to resilient, affordable infrastructure and those forced to rely on informal, high-cost alternatives. By analyzing the intersection of governance, economics, and climate resilience, we examine how urban services—from water and sanitation to transport and energy—dictate the economic mobility of billions.

BLOG SECTION: Critical Perspectives

1. Bridging the Gap: Resilient Infrastructure for All

About 1 in 3 urban residents globally lack access to essential services. This is not merely a logistical failure; it is a systemic economic barrier.

Expanded Explanation:

Infrastructure is often viewed as a static "thing"—pipes in the ground or wires in the air. However, in the context of urban economics, infrastructure is a **dynamic flow of opportunity**. When a city fails to extend its formal grid to the periphery, it creates a physical boundary for economic growth. For residents in underserved areas, the lack of a formal connection means they must spend a significant portion of their day simply securing basic needs, which subtracts from time that could be spent on education or labor.

Furthermore, infrastructure must be **resilient**. In a changing climate, infrastructure that is not built to withstand shocks is a liability. For a family living on the edge of poverty, a single burst pipe or a flooded road can mean the difference between a productive week and a complete loss of household assets.

Economic Analysis:

- **Inequality:** The "poverty premium" describes how the poor pay more for less. In cities like Nairobi or Manila, residents of informal settlements often pay 5 to 10 times more per liter of water to private vendors than those with municipal taps.
- **Environmental Costs:** Without resilient drainage, informal settlements suffer the highest damage costs from climate events, acting as an "inequality multiplier."

2. The Digital and Data Evolution of City Utilities

Modern utilities are no longer just about pipes and wires; they are about data flows.

Expanded Explanation:

We are entering an era where "bits" are as important as "bricks." Smart utility management—utilizing the Internet of Things (IoT) and AI—allows cities to monitor infrastructure health in real-time. For instance, sensors can detect a leak in a water main miles away before it causes a sinkhole. This shift is critical for cities with shrinking budgets, as it moves maintenance from a **reactive** model (fixing things after they break) to a **predictive** model.

However, a digital divide can exacerbate the service divide. If sensors and smart meters are only installed in wealthy, "formal" districts, urban planners remain "blind" to the needs of the informal city. Data-driven governance must be inclusive to be effective.

Economic Analysis:

- **Efficiency:** Reducing "non-revenue water" (water lost to leaks or theft) can save municipal utilities millions of dollars, which can then be reinvested in expansion.
- **Productivity:** Reliable electricity and internet are prerequisites for the modern "gig" and service-based economies, which are the primary drivers of urban GDP.

3. Structural Transformations to Solve Urban Inequality

Inequality is "baked in" to the physical layout of many cities.

Expanded Explanation:

Urban inequality is a **spatial trap**. When housing is disconnected from transit, low-income workers are forced to spend upwards of 30% of their income on commuting. This is a massive drain on the urban economy. True transformation requires "Cross-Sector Reform"—the realization that you cannot solve the housing crisis without simultaneously solving the transit and sanitation crises.

Integrated urban planning views the city as a single organism. For example, building a high-capacity bus rapid transit (BRT) line into an informal settlement doesn't just move people; it brings that settlement into the formal labor market, increasing the city's total productive capacity.

Economic Analysis:

- **Labor Access:** Improving transit density directly increases the "matching efficiency" of the labor market, allowing workers to find jobs that best suit their skills.
- **Public Spending:** Integrated solutions (like multi-purpose drainage that doubles as public parks) provide higher "social returns on investment" than single-use projects.

Research Synthesis

Core Water & Governance Papers

- **Gozun & De Castro (2026):** Their study in the Philippines highlights that **reliability** is the primary driver of household satisfaction. It isn't enough to have a tap; if water only flows for two hours a day, the household still operates in a state of "water poverty."
- **Bibri & Krogstie (2024):** They argue that digital transformation is the "glue" of the sustainable city. They advocate for a **Symbiotic Relationship** where data informs governance, which in turn improves service delivery.
- **Gerlach et al. (2023):** This research identifies the **Enabling Environment**. Technical engineering is easy; what is hard—and more important—are the legal and financial frameworks that allow a utility to be self-sustaining and autonomous from political cycles.

Thematic Research: Challenges & Solutions

Theme	Key Finding	Long-Term Economic Effect
Traffic	Machine learning (Rahman et al.) predicts congestion patterns.	Minimizes lost "man-hours" and lowers urban carbon emissions.
Flooding	Informal settlements (Santos et al.) are often in high-risk zones.	Prevents the "decapitalization" of poor households after storms.
Housing	Adequacy requires services, not just walls (Turok et al.).	Proper housing acts as the "anchor" for health and economic stability.
Waste	Taxes and tech drive waste-to-energy (Imran et al.).	Turns a public health liability into a resource for the circular economy.

SYNTHESIS: The Future of Our Cities

Conclusion & Policy Outlook

To close the urban services divide, we must move beyond the "grid vs. off-grid" mentality. The future of the resilient city lies in **Technical and Governance Hybridization**—where decentralized solutions (like solar mini-grids or community-led sanitation) are integrated into the formal municipal plan.

Key Policy Recommendations:

1. **Incentivize Inclusion:** Change utility regulations so that companies are rewarded for extending services to informal areas, rather than penalized for the "risk."
2. **Climate-Proof the Periphery:** Prioritize infrastructure spending in low-income, high-risk geographic areas to prevent climate-driven poverty traps.
3. **The Compact City Model:** Encourage higher density with mixed-use development to make the per-capita cost of service delivery lower and more sustainable.

Website Enhancement Suggestions:

- **Interactive Maps:** A "Service Layer" map where users can toggle between water access, transport lines, and income levels to see the "divide" visually.
- **Utility Dashboard Simulation:** A small interactive widget where students can try to balance a city's budget while choosing between "Repairing Old Pipes" vs. "Extending to New Slums."
- **Infographic:** A "Circular Urban Metabolism" chart showing how waste from one sector becomes an input for another.

Videos/Summaries

Choices

1. https://ocw.tudelft.nl/course-lectures/introduction-urban-services-infrastructures/
2. https://unhabitat.org/knowledge/global-urban-lectures
3. https://youtu.be/toEQZapHYLA?si=uJWbk0D41pOlzUqH
4. https://youtu.be/2kIS71JA1Lk?si=BqsjBsqEKbOMeZm6
5. https://youtu.be/Q1NLATszskU?si=4Y8VrpPAFAIx7op_
6. https://www.wri.org/insights/transformations-to-solve-urban-inequality
7. Video Shorts: Urbanization – The Human Imprint
8. https://archive.metabolismofcities.org/videos/43-what-is-urban-metabolism
9. https://archive.metabolismofcities.org/videos/45-participatory-urban-metabolism-in-medellin-colombia
10. https://smartcity.go.kr/2017/10/03/smart-cities-building-for-the-cities-of-tomorrow-documentary-2015/
11. https://www.worldbank.org/en/news/video/2017/04/19/cities-are-where-the-future-is-being-built
12. https://unhabitat.lk/multimedia/state-of-sri-lankan-cities-report-multimedia/video-urban-sprawl-and-compact-cities/
13. https://youtu.be/0SoKj5Gxnnk?si=8FyWVB96UjgKFRgY
14. https://youtu.be/tgYOOPMQmc4?si=dA2Cqe3fs_ErdGM0
15. https://youtu.be/ni3teWmnOR0?si=qtt_XJnh5PxovGrg

(you can put these in any order u want, ray :>>)

Video 1: AMS.URB1X 2016 1.4.1 Introduction to Urban services & Infrastructure p… /https://ocw.tudelft.nl/course-lectures/introduction-urban-services-infrastructures/

- This video shows how urban infrastructure acts as the foundation of sustainable and equitable urban development.
- It discusses how infrastructure is more than just physical structures such as roads or pipes. It includes the systems and services that support daily life like the distribution of energy and water. Using the UN Habitat City Prosperity Index, the video shows how prosperity depends on several factors including infrastructure, equity, productivity, and environmental sustainability.
- It also emphasizes the need for cities to adapt to climate change through mitigation and adaptation strategies, especially in sectors like transport and energy. It emphasizes shifting from a linear economy, which creates waste, to a circular economy focused on reuse and sustainability in order to achieve this.

Video 2: Urban Infrastructures & Services - 1

- Overall, the video lecture emphasizes that infrastructure is not just about construction, but about delivering reliable and equitable services that support urban development.
- It explains the difference between physical systems like roads, water networks, and power grids and the services they provide, such as better transport, clean water, and energy. It also highlights how planners and local governments are responsible for creating systems that improve quality of life, economic opportunity, and access to essential services.

Video 3: Urban Infrastructure - How a City Designs its Utilities and Public Services

- This video discusses how infrastructure in cities refers to the systems that keep urban life running, including utilities like water, electricity, housing, and etc. These things determine how a city's quality of life is through how efficiently people can work and their general wellbeing.
- It also talks about how strong infrastructure improves economic growth and daily convenience, while weak infrastructure can create inequalities and inefficiencies. Cities usually design infrastructure to adapt to factors like their geography, population, and economy. Examples of which include Perth relying on highways and ports for resource-driven growth, and Amsterdam using canals, cycling systems, and flood management to adapt to low-lying land.

Video 4: AMS.URB1X 2016 3.4.1 Changes in urban utility infrastructure

- This video covers the evolution of urban utility infrastructure in terms of water and energy systems. These systems have evolved through three main phases: early localized networks run by private entrepreneurs, mid-20th century nationalized public utilities focused on universal access and public health, and the modern shift toward marketized and decentralized systems with multiple providers. Each phase changed how services were delivered, from small city-based systems to large interconnected national grids, and now toward more flexible and distributed infrastructures.

Video 5: What is Urban Metabolism?

- This video focuses on urban metabolism, which treats cities like systems where resources flow inwards and waste flows out. Cities usually still follow a linear model that produces large amounts of waste, while a circular approach reuses materials, such as converting wastewater into energy or fertilizer, which minimizes the amount of waste that comes out. Shifting toward this circular system can reduce costs, emissions, and resource dependence while improving long-term urban resilience and equity.

- This is relevant because cities have become the main form of human settlement, consuming most global resources and producing the majority of emissions. Following this concept helps urban services and utilities become more efficient and sustainable.

Video 6: Why Unequal Cities Matter: Closing the Urban Services Divide to Benefit …

Summary

The video addresses the **urban services divide**, highlighting the stark disparities in access to essential services within cities worldwide. It emphasizes that **not all urban residents experience city life equally**, with daily activities being straightforward for some but a significant challenge for others. This divide is particularly pronounced in low-income countries, where a substantial portion of the urban population lacks consistent access to basic services.

Key Insights

- Over **one billion people globally** face challenges due to inadequate urban services.
- In **low-income countries**, approximately **two-thirds of urban residents are underserved**.
- Essential services such as **water supply** may be available only intermittently, sometimes **just a few hours per week**.
- **Access to employment** often involves **arduous and unsafe journeys**, adding to the burden.
- The urban services divide leads to:
    - **High time costs** for basic needs.
    - **Self-provision practices**, where individuals must find their own solutions.
    - **Unhealthy and polluting practices**, adversely affecting both human health and the environment.
- The **cumulative human and environmental costs** of these disparities negatively impact not only underserved populations but society as a whole.
- **Closing the urban services divide yields benefits for everyone**, emphasizing the importance of equitable service provision.

Video 7: Why Cities Are Where They Are

Summary

This Wendover Productions video explores the geographic and economic principles behind the location and spacing of towns and cities, emphasizing how natural features and human needs shape urban development.

Key Insights on Town Spacing and Development

- **Cumberland Valley Towns Example:** Six small Pennsylvania towns—Greencastle, Chambersburg, Shippensburg, Newville, Carlisle, and Mechanicsburg—are spaced approximately **10 miles apart**, with deviations of no more than a mile. This spacing is **not coincidental or planned**, but rather an outcome of practical human mobility and economic efficiency before the advent of cars.
- **Walking Distance and Efficiency:**
    - Before cars, most people walked everywhere.
    - A **5-mile radius** (half the distance between towns) represented the maximum comfortable walking distance for daily commerce.
    - This distance allowed sufficient time for buying or selling goods in a central market, making **10 miles the ideal distance between towns** for efficient rural land use.
- **Spheres of Influence and Services:**
    - Towns serve a local population that sustains basic shops such as grocery stores, pharmacies, banks, and restaurants.
    - More specialized services (e.g., mechanics) require larger populations and hence are located in bigger towns or cities with larger spheres of influence.
    - Cities have the largest spheres of influence, offering both common and highly specialized services (e.g., luxury car dealerships, brain surgery centers, airports).
- **Example of Service Density in Cities:**
    - Starbucks locations in lower Manhattan are about **600 feet apart**, each serving roughly **6,000 people**.

- Apple Stores require about **500,000 people** within a 10-mile radius to be sustainable, as illustrated by the spacing between stores in Connecticut.
- Income levels affect these numbers, with wealthier areas requiring fewer people per store.

Conclusions

- **Urban spacing and growth are fundamentally tied to human mobility, economic sustainability, and geographic features.**
- **Before modern transportation, walking distances dictated town placement.**
- **Spheres of influence grow with specialization and population density, explaining the hierarchy from towns to cities.**
- **Water bodies, natural resources, and terrain strongly influence urban location.**
- **The historical shape and climate consistency of continents have shaped the distribution of civilizations and cities.**
- **South-central Asia, particularly Dhaka, represents an optimal urban location from a purely geographic and demographic standpoint.**

Video 8: What is Urban Metabolism?

Summary of Urban Metabolism and Sustainable Cities

Since 2007, more people live in urban areas than in rural ones for the first time in history. Cities currently consume **75% of Earth's resources** and contribute to **60-80% of global greenhouse gas emissions**. With rapid urbanization, it is projected that by 2050, over **two-thirds of the world's population will reside in cities**. This significant growth places immense pressure on Earth's limited resources, challenging the capacity of cities to sustain themselves while maintaining quality of life, especially for vulnerable populations.

Key Challenges

- At current consumption rates, humanity would require the equivalent of **1.6 Earths** to meet resource needs and waste absorption.

- However, Earth provides only **one planet's worth of resources**, posing a limit that necessitates more efficient use.
- Cities face the challenge of delivering services that are **resource-efficient, climate-friendly, resilient, and equitable**.

Urban Metabolism Concept

Urban metabolism is a framework that views a city as a living organism:

- **Definition:** It tracks the flow of resources into a city and waste outputs out of it.
- Helps identify how cities consume resources and generate wastes.
- Provides insights into improving resource efficiency and reducing environmental impacts.

Linear vs. Circular Metabolism

- **Linear metabolism** is the dominant pattern in many cities, where resources enter the city and leave as solid, liquid, or airborne waste with little reuse.
- For sustainability, cities must transition to a **circular metabolism** by closing resource loops and reusing wastes as valuable resources.

Examples of Circular Metabolism

- Wastewater treatment plants can serve multiple functions:
  - Cleaning water
  - Capturing methane for energy
  - Recovering nutrients to enrich urban soils
- This recirculation reduces dependency on external resource inputs and minimizes waste outputs.

Benefits of Urban Metabolism Studies

- Identifies **infrastructure and planning interventions** that optimize resource use.
- Helps cities save money over time through improved efficiency.
- Contributes to **greenhouse gas reduction**, notably CO and methane emissions.
- Enables city stakeholders to:
  - Monitor resource consumption
  - Learn from other cities' experiences
  - Educate decision-makers on sustainable infrastructure options

- Improve resource efficiency in buildings and operations

Conclusions

- **Urban growth demands a shift from linear to circular resource use** to ensure sustainability.
- Employing urban metabolism principles supports **resource conservation, climate mitigation, and social equity**.
- Cities must innovate in infrastructure and policy to **close resource loops and reduce waste**.
- Stakeholder engagement and knowledge sharing are critical to advancing sustainable urban metabolism strategies.

Video 9: Participatory Urban Metabolism in Medellín, Colombia

Summary

The video content revolves around themes of **environmental awareness, community involvement, and sustainable living practices**, conveyed through a combination of music and spoken commentary. It underscores the importance of **care and responsibility toward the environment**, while highlighting specific initiatives related to **recycling and organic waste management** within a local community.

Key Insights

- **Environmental care is a collective responsibility**, emphasized metaphorically as “taking care of the house with the pieces,” which suggests that every small action contributes to the overall well-being of the planet.
- The video warns against **indifference, impatience, unconsciousness, and violence**, which hinder progress in environmental stewardship.
- There is a call for **honesty, awareness, and patience** in the community’s approach to sustainability and coexistence.

- **Recycling plays a crucial role** in this environmental effort, but only **three types of plastics** are recyclable:
  - Bottles
  - Bags
  - Hygiene product packaging (e.g., shampoo containers)
- The community has established a **recycling and waste management system**:
  - A point of collection for **used cooking oil**, partnered with a local company and the church in the Boston Park area.
  - A **biodigester system for organic waste**, currently used by around **seven neighboring households** out of an initial group of about ten to fifteen.
- The initiative has been ongoing for approximately **two and a half years**, demonstrating sustained commitment.
- The narrative encourages **environmental action as more than a trend**, promoting it as a genuine and necessary lifestyle change.

Core Concepts

- **Environmental stewardship** requires collective vigilance and ongoing effort.
- **Patience, consciousness, and community cooperation** are key for successful sustainable living.
- Recycling and organic waste management are practical steps to reduce environmental impact.
- Local partnerships and community-led initiatives enhance resource management and promote sustainability.

Conclusions

- This video presents a **holistic approach to environmental care** that combines cultural expression (music) with practical community actions.
- **Sustainability is framed as a continuous and honest commitment**, not a passing trend.
- The community's success in establishing **plastic recycling and biodigester use** exemplifies effective grassroots environmental management.

- **Awareness of recyclable materials and responsible waste disposal** is critical to these efforts.
- The narrative urges vigilance against complacency and advocates for **active participation in caring for the environment**.

Video 10: Smart Cities - Building for the Cities of Tomorrow (Documentary, 2015)

Summary of Video Content: Sustainable Urban Development and Future Cities

The video explores the rapid urbanization of the global population, noting that **50% of people now live in cities**, compared to only 10% at the start of the 20th century. This shift presents both challenges—such as **scarcity of space, pollution, noise, and social tensions**—and opportunities for **resource efficiency, energy optimization, and innovative infrastructure**. The overarching theme is the urgent need to create **sustainable, energy-efficient urban environments** that can support future generations.

Key Highlights and Insights

- **Sustainable Building as a Technological Key:**
  Austrian researchers and architects are pioneers in sustainable construction, exemplified by the **Sheikh Zayed Desert Learning Center** in Abu Dhabi. This project integrates **traditional architectural elements with advanced technologies** to create a building that remains cool despite outside temperatures of up to 50°C.
- **Desert Adaptation Technologies:**
  - One-third of the museum is built underground for natural cooling.
  - Ventilation air is pre-cooled via an 800-meter underground pipe system combined with a geothermal heat exchanger.
  - Solar-powered air conditioning uses solar thermal collectors driving chillers for cooling.

- Fully automated electronic control systems monitor and maintain building functions remotely.

  These approaches yield annual electricity savings of **25 megawatt-hours**.

- **The UAE's Sustainability Challenge and Vision:**
  - The UAE, an oil-rich country with a high carbon footprint, is transitioning toward renewable energy and sustainable urban planning.
  - Projects like **Masdar City** aim to create a **CO2-neutral urban oasis** in the desert for 40,000 residents and 50,000 commuters, featuring low-rise buildings designed to reduce heat and optimize wind flow via orientation.
  - Masdar City is also a research hub with over 490 students from 60 countries focused on renewable energy and sustainability.
- **Innovative Urban Planning Concepts:**
  - Two urban development trends identified:
    - **Mega-cities divided into compact, pedestrian-friendly neighborhoods interconnected on a larger scale.**
    - **City marketing to position themselves globally with distinct identities.**
  - The Austrian city of Graz is developing **energy self-sufficient urban quarters** that integrate industry, residential, and green spaces, with examples like the **Rining House district**, where waste heat from a steel mill heats 10,000 homes.
- **Plus Energy Buildings and Competitions:**
  - Students at Vienna University of Technology designed a **Plus Energy House** (generating more energy annually than it consumes) for the **Solar Decathlon** competition in California.
  - The Austrian team won first place with **952 out of 1000 points**, highlighting Austria's leadership in sustainable building innovation.
- **Retrofitting Existing Buildings:**

    - The university's chemistry building in Vienna is being renovated into the world's first **plus energy high-rise**, featuring a **320 kW peak photovoltaic system** on the roof and façade.
    - Power consumption reduction of about **90%** is achieved by optimizing or replacing over 9,300 electrical components (lighting, computers, etc.).
- **Smart Grids and Renewable Energy Integration:**
    - In **Kristin Dorf, Austria**, a pilot smart grid manages 43 photovoltaic systems (total peak output 200 kW) and coordinates energy use between homes, businesses, and electric cars serving as buffer storage.
    - Smart regulators balance intermittent renewable energy supply with demand, enhancing grid stability.
    - Salzburg is expanding this model to overhaul its entire power system, paralleling the telecommunications revolution.
- **Social and Mobility Innovations:**
    - Projects like Graz's **Rosa Zukunft (Rosy Future)** combine renewable energy with social concepts to foster community and reduce car dependency through walkability and bike-friendly infrastructure.
    - Despite success in energy savings (~15% reduction in first year), adoption of alternative mobility (car sharing, EVs) remains a challenge.

Video 11: Cities Are Where the Future Is Being Built

Cities, which generate over 80% of global GDP are drivers of economic growth, they are the anchors of development in a shifting global landscape. Matchmakers connecting people to jobs. It is estimated that 66% of people in the world will live in cities by 2050. But cities are under pressure from global economic slowdowns, natural disasters, climate change, and the need to improve basic services for the urban poor. There are some ideas to make the cities more resilient, inclusive, productive, livable, sustainable (construction, food services, etc.).

Video 12: Urban Sprawl and Compact Cities video

Urban Sprawl is one of the key challenges facing Sri Lankan cities today. Urban Sprawl involves rapid land consumption as cities expand and swallow up surrounding rural areas.

Sprawl development = uncoordinated, low density urban expansion

Sprawl negatively impacts urban mobility because commuters have to travel greater distances

Sprawl threatens our green spaces, ecological habitats and increases vulnerability to natural disasters.

Sprawl is also expensive in increasing the cost of public services.

Compact cities are high-density and only cover a small area. Across the globe, urban planners are striving to promote compact cities as an antidote to sprawl.

Compact urban growth promotes environmental sustainability, encourages non motorized mobility, enables widened access to key services, helps retain our green cover, contributes to the health and wellbeing of urban residents

Video 13: URBAN PLANNING AND CITY GOVERNANCE

Summary

The video discusses the concept that living in densely populated cities can be more environmentally friendly. It highlights the principles of sustainable urban planning inspired by the ancient Indian concepts of **Satyam, Shivam, and Sundaram**—meaning sustainable, efficient, and aesthetic.

Key Insights

- **Ecological Sustainability**: Ensuring natural resources like water and green cover match population needs.
- **Economic Sustainability**: Evaluating the long-term costs of supporting a population.
- **Political Sustainability**: Using governance structures to maintain urban settlements.

- **Efficiency in Urban Planning**:
  - Logical zoning considering natural features and population needs.
  - Comprehensive building bylaws for construction, transportation, and public amenities.
  - Efficient public services such as health care, waste disposal, and disaster management.
- **Aesthetic Considerations**: Designing cities that reflect cultural heritage and architectural beauty.

Conclusion

Urban planning is a complex discipline integrating arts, sciences, and governance. The video emphasizes that cities should prioritize people over cars, reflecting the cultural and civic sense of their inhabitants.

Video 14: Can AI Fix Traffic?

This video explores Google's new AI-powered traffic management system, **Project Greenlight**, designed to reduce traffic congestion in urban areas of the US. The system aims to improve traffic flow by analyzing data from Google Maps to identify inefficiencies at intersections and recommend adjustments to traffic light timings. This technology offers a cost-effective solution for cities, potentially reducing emissions and improving traffic efficiency without extensive hardware investments.

Key Insights

- **Cost-Effective Solution**: Project Greenlight is an inexpensive alternative to traditional traffic management methods, allowing cities to optimize traffic flow without significant financial investment.
- **Emissions Reduction**: By minimizing stop-and-go traffic, the system can potentially reduce emissions at intersections by up to 10%.
- **Implementation**: The system has been successfully tested in Seattle, demonstrating its practicality and effectiveness in real-world settings.
- **Wide Applicability**: Suitable for cities of various sizes due to its low cost and minimal hardware requirements.

- **Potential Drawbacks**: There is a risk of inducing demand, where improved traffic conditions could encourage more driving, potentially negating the benefits by increasing congestion.

Video 15: Singapore’s AI traffic management system

Summary of Singapore’s AI Traffic Management System

Singapore has developed a cutting-edge artificial intelligence (AI) traffic management system that significantly reduces traffic congestion, setting a global example in urban mobility. Unlike most cities where gridlock consumes countless hours annually, Singaporean drivers experience minimal delays due to a sophisticated, data-driven approach.

Key Insights

- **Traffic Congestion Comparison:**
    - Average American loses 54 hours/year in traffic.
    - Londoners lose 148 hours/year.
    - Singaporeans lose only 25 hours/year.
- This dramatic reduction results from a **$2.4 billion investment** in an **Intelligent Transport system (IT)** that leverages AI and machine learning.

Tasks and Output Matrix

Member	Tasks/s	Output/s
1. John Raymond Tugadi	- Website coding	
2. Jonathan Margate Verceles	- Website coding & Content Research - Additional academic/journal papers - Summarization of academic/journal papers	
3. Julion Miguel Tin	- Site layout/ Hierarchy - Documentation - Website Coding	
4. Miguel Lorenzo Clemente	- Content Research - Video Content Summarization	
5. Miguel Jacinto Rosario	- Content Research	
6. Ramon Miguel Munarriz	- Content Research	
7. Pia Ballesteros	- Site Content/Formatting	

Documentation

Submission Requirements

Minimum content requirements. You can get sample materials from the Internet. You may also use your own materials.

- Five blogs
- Five academic/journal papers
- Five videos
- 10 photos with short caption each
- Relevant links

Outputs to be Submitted (Deadline: 22 May 2026)

- Working link of the website with the contents: https://econ-state-of-urban-services-provis.vercel.app/?fbclid=IwY2xjawRtN8lleHRuA2FlbQIxMQBzcnRjBmFwcF9pZAEwAAEeG176uThnQUPafuK9AUhBSS4xdRtwSKyQcyRVMGoCG37EV8lps6lDHm0BBGw_aem_E1L7cKL887OQGeEDga8AVA
- Source Code:
- Demo of how the website works (screen recording):
- Documentation: ECN100 Research Doc / https://docs.google.com/document/d/1Mz1XASagd7zSSBzy7iRJmRVSvKe9ugcLZf0mLvrlkV4/edit?tab=t.xkfymvy6h8g7
- Member's Contribution Matrix (see template below): ECN100 Research Doc / https://docs.google.com/document/d/1Mz1XASagd7zSSBzy7iRJmRVSvKe9ugcLZf0mLvrlkV4/edit?tab=t.aaniwxqybs53

ECN100 Research Folder

Paper Summaries

Five academic/journal papers

1. Gozun, B. C., & De Castro, J. M. (2026). Household perspectives on urban water service delivery: Implications for policy and practice in the Philippines. *Public Works Management & Policy*. Advance online publication. https://doi.org/10.1177/1087724X261439281
    a. Note: Can't access the pdf

2. Das, D. K. (2024). Exploring the Symbiotic Relationship between Digital Transformation, Infrastructure, Service Delivery, and Governance for Smart Sustainable Cities. Smart Cities, 7(2), 806-835. https://doi.org/10.3390/smartcities7020034
    a. Tentative paragraph:
        i. This paper explores how digital transformation, infrastructure, service delivery, and governance operate in a symbiotic relationship to shape the provision of urban services and drive the emergence of smart sustainable cities. The author argues that digital transformation—integrating technologies such as AI, IoT, and big data—serves as a catalyst, making physical and virtual infrastructure more efficient, resilient, and adaptive. This upgraded infrastructure, in turn, becomes the backbone for seamless, high-quality service delivery in areas like water, energy, transport, healthcare, and education. Governance ensures that these services remain aligned with citizens' evolving needs through transparency, accountability, and data-driven participatory decision-making. The synergy is mutual: digital tools enhance governance, governance enables better infrastructure planning, and well-designed infrastructure facilitates responsive service provision. Case studies from Singapore's Smart Nation Initiative, Estonia's X-Road platform, India's Aadhaar and UPI systems, the UK's Gov.uk portal, and Rwanda's digital land registry illustrate how these four pillars reinforce each other to improve service accessibility, reduce corruption, and promote transparency. The study concludes that only by recognizing and leveraging these interdependent linkages can cities transcend mere technological upgrades to become truly smart, sustainable, and inclusive, ensuring that urban services are efficiently delivered to all citizens in a transparent and accountable manner.
    b. V2 paragraph:
        i. This paper discusses how digital transformation, infrastructure, service delivery, and governance work together to improve urban services and help develop smart and sustainable cities. Dillip Kumar Das, the author,

explains that digital transformation, through technologies like AI, IoT, and big data, acts as a driving force that makes both physical and digital infrastructure more efficient, reliable, and adaptable. This improved infrastructure then supports better service delivery in sectors such as water, energy, transportation, healthcare, and education. Governance plays an important role by making sure these services meet the changing needs of citizens through transparency, accountability, and data-driven decision-making. These elements are interconnected because digital tools improve governance, governance supports better infrastructure planning, and strong infrastructure allows services to be delivered more effectively. Examples from Singapore's Smart Nation Initiative, Estonia's X-Road platform, India's Aadhaar and UPI systems, the UK's Gov.uk portal, and Rwanda's digital land registry show how these four areas strengthen one another to improve accessibility, reduce corruption, and increase transparency. The study concludes that cities can only become truly smart, sustainable, and inclusive if they recognize and maximize the connection between these factors, ensuring that services are delivered efficiently and fairly to all citizens.

c. Alternative tentative paragraph:
    i. The paper by Dillip Kumar Das examines the state of urban service provision through the lens of smart sustainable cities, emphasizing the interconnected roles of infrastructure, governance, digital transformation, and service delivery. The study argues that effective urban services—such as transportation, healthcare, water supply, sanitation, education, and digital connectivity—depend on strong and well-planned infrastructure supported by modern technologies like Artificial Intelligence (AI), the Internet of Things (IoT), and digital platforms. It highlights that cities today face growing challenges related to population growth, environmental sustainability, accessibility, and governance, making efficient service delivery more critical than ever. The paper explains that digital transformation improves urban services by enabling faster communication, data-driven decision-making, real-time monitoring, and more responsive governance systems. Examples from countries such as Singapore, Estonia, India, the United Kingdom, and Rwanda demonstrate how digital governance and smart infrastructure can improve public services, increase transparency, reduce corruption, and enhance citizen participation. However, the study also notes challenges including cybersecurity risks, data privacy concerns, unequal access to technology, and fragmented urban development. Overall, the paper concludes that sustainable and efficient urban service provision requires a symbiotic relationship among digital innovation, infrastructure development, governance, and citizen-centered service delivery to create smarter, more inclusive, and resilient cities for the future.

3. Trimmer JT, Qureshi H, Otoo M, Delaire C (2023) The enabling environment for citywide water service provision: Insights from six successful cities. PLOS Water 2(6): e0000071. https://doi.org/10.1371/journal.pwat.0000071
    a. Tentative Paragraph:
        i. This study investigates the enabling environment for inclusive citywide piped water services by analyzing six cities in low- and middle-income countries—Abidjan, Ahmedabad, Bangkok, Cairo, Phnom Penh, and Porto Alegre—that achieved high levels of piped water access and service quality, even in low-income areas. Using a modified social-ecological systems framework, the authors identify three distinct pathways of progress: utility-driven (where autonomous, champion-led utilities instituted efficiency reforms and public engagement), regulator-supported (where independent regulators monitored performance and coordinated investment), and municipality-driven (where local government integrated water provision with slum upgrading and pro-poor mandates). Across these pathways, twelve common characteristics of the enabling environment emerge, including clear performance indicators, staff incentives and anti-corruption measures, pro-poor connection subsidies coupled with the removal of land tenure requirements, community mobilization and participatory decision-making, robust metering to reduce non-revenue water, financial sustainability through tariffs or property taxes, alignment with donor priorities, democratic and participatory political practices, shifts in public mindsets recognizing informal residents' rights, and leveraging environmental crises to drive water security reforms. The findings highlight that a well-functioning, accountable water service provider is a prerequisite for inclusive delivery, but explicit pro-poor strategies—such as bypassing land tenure barriers, financing mechanisms that reach the poorest, and institutional mandates to serve informal settlements—are essential to close equity gaps. These insights offer transferable lessons for cities striving to improve urban water provision, demonstrating that success depends not on a single model but on context-sensitive combinations of governance, finance, and community engagement that together make universal access to safe water an achievable goal.
    b. V2 Paragraph:
        i. This study examines how some cities in low- and middle-income countries successfully provided inclusive piped water services, even in low-income communities. By analyzing cities such as Abidjan, Ahmedabad, Bangkok, Cairo, Phnom Penh, and Porto Alegre, the authors identified three main approaches: utility-driven, regulator-supported, and municipality-driven systems. Despite their differences, these cities shared common factors like strong accountability, anti-corruption measures, community participation, financial sustainability, and policies that supported low-income residents, such as subsidies and removing land ownership

requirements for water access. The study emphasizes that effective and accountable water providers are necessary, but targeted pro-poor strategies are equally important to ensure fair access to safe water. Overall, the findings show that there is no single solution for improving urban water services, as success depends on combining governance, financing, and community involvement based on the city's specific context.

c. Alternative Tentative Paragraph
    i. The study by John T. Trimmer explores how cities in developing countries can improve inclusive and sustainable urban water service provision by examining six successful cities across Africa, Asia, and South America: Abidjan, Ahmedabad, Bangkok, Cairo, Phnom Penh, and Porto Alegre. The paper emphasizes that equitable access to safe drinking water remains a major challenge in rapidly urbanizing cities, especially for low-income communities that often lack reliable infrastructure and formal service connections. Using a comparative case study approach, the researchers identified key factors that create an "enabling environment" for effective urban water services, including strong governance, clear regulations, financial sustainability, community participation, and institutional accountability. The study found that successful cities implemented pro-poor policies such as subsidized water connections, flexible payment systems, removal of land ownership requirements for access to piped water, and participatory planning involving local residents. It also highlighted the importance of efficient utilities, transparent performance monitoring, customer feedback systems, and investment in modern infrastructure to improve reliability and reduce water losses. Additionally, political commitment, international funding, environmental management, and digital monitoring systems played important roles in strengthening water service delivery. The paper concludes that while different cities followed different development paths—utility-driven, regulator-supported, or municipality-driven—all successful cases shared a common focus on inclusive governance, financial efficiency, and policies designed to ensure that even the poorest urban residents gain access to safe and affordable water services.

4. Moretto, L., Faldi, G., Ranzato, M., & Rosati, F. (2023). Coproduced urban water services: When technical and governance hybridisation go hand in hand. *Frontiers in Sustainable Cities, 4*, 969755. https://doi.org/10.3389/frsc.2022.969755
    a. Tentative Paragraph:
        i. This study examines the evolution of water and sanitation service coproduction in four cities of the Global South—Hanoi, Dar es Salaam, Cochabamba, and Addis Ababa—through the lens of sociotechnical hybridisation, focusing on the interplay between infrastructure

configurations and governance arrangements. Drawing on a conceptual distinction between hybridity as a fixed form and hybridisation as an ongoing process, the analysis shows that coproduced services continuously adapt through complementary technologies (e.g., household storage tanks, tertiary pipes, in-house connections that extend or fill gaps in centralised networks) and concurrent systems (e.g., community-managed decentralised networks operating alongside municipal ones). Governance hybridisation unfolds through the emergence of new intermediaries—such as water committees, resident councils (TDPs in Hanoi, OTBs in Cochabamba), and private water resellers—whose roles often blur the boundaries between state and non-state actors, yet generally maintain concordant institutional multiplicity rather than corrosive overlaps. The cases reveal that group coproduction, once institutionalised through state-facilitated committees managing shared fountains or decentralised networks, has increasingly given way to individual coproduction (private connections, storage, and resale) as municipal networks have expanded and residents' aspirations for in-house connections have grown. Drivers of these shifts include changes in regulatory frameworks, municipal infrastructure strategies favouring revenue water and universal coverage, environmental pressures (e.g., groundwater depletion in Cochabamba), and the progressive retreat or reconfiguration of state responsibilities. The findings underscore that hybridisation is a pragmatic, problem-solving process rather than a deliberate democratic deepening; while it improves access, it often fails to resolve underlying problems of equity and sustainability, as seen in the exclusion of newer residents in Cochabamba or the dissolution of collective management structures in Addis Ababa and Dar es Salaam. The study thus highlights that coproduction's potential to deliver sustainable inclusive services depends on recognising hybridisation as a dynamic tension between multiple actors and technologies, not as a stable endpoint.

b. V2 Paragraph:
    i. This study explores how water and sanitation services developed in four Global South cities—Hanoi, Dar es Salaam, Cochabamba, and Addis Ababa—by looking at how infrastructure and governance continuously adapt over time. The authors explain that coproduced services evolve through both shared and individual solutions, such as household storage tanks, private water connections, community-managed systems, and water resale. New groups like water committees, resident councils, and private resellers also emerged, creating cooperation between governments and local communities. Over time, however, collective systems gradually shifted toward more individual-based solutions as city networks expanded and residents preferred direct household connections. These changes were influenced by government policies,

environmental pressures, and changing state responsibilities. While hybrid systems helped improve access to water and sanitation, the study points out that they did not fully solve issues of inequality and sustainability. Overall, the research shows that successful service delivery depends on flexible cooperation between different actors and technologies rather than relying on one fixed system.

c. Alternative Tentative Paragraph:
    i. The study by Luisa Moretto examines how urban water services in rapidly growing cities of the Global South are increasingly shaped through “coproduction,” a process in which governments, communities, and private actors work together to provide water services where centralized systems are insufficient. Focusing on four cities—Hanoi, Dar es Salaam, Cochabamba, and Addis Ababa—the paper explains that rapid urbanization, poverty, informal settlements, and weak infrastructure often prevent cities from delivering universal and reliable water access through traditional centralized networks alone. As a result, communities have developed hybrid systems that combine formal infrastructure with local practices such as community-managed wells, shared fountains, water storage tanks, household pipes, and informal water reselling. The research highlights that urban service provision is not static but continuously evolves through “technical and governance hybridization,” where both infrastructure systems and institutional responsibilities change over time. In many cases, governments still manage the main water networks while communities or local groups handle maintenance, distribution, or storage at the neighborhood level. The paper also discusses how these arrangements can improve accessibility and affordability for low-income residents, yet they may also create inequalities, financial pressures, and governance challenges when water resources become limited or when services become privatized. Ultimately, the study concludes that effective urban water service provision requires flexible and collaborative governance systems that adapt to changing urban conditions while balancing state responsibility, community participation, and sustainable infrastructure development to ensure more inclusive access to essential urban services.

5. La Vigna, F. (2022). Urban groundwater issues and resource management, and their roles in the resilience of cities. *Hydrogeology Journal*, *30*(6), 1657-1683. https://doi.org/10.1007/s10040-022-02517-1
    a. Tentative Paragraph:
        i. This review examines the critical but often overlooked role of urban groundwater in sustaining city services and enhancing resilience, drawing on examples from over 70 cities worldwide. It highlights how groundwater—though hidden beneath pavement and infrastructure—provides essential services such as drinking water supply,

irrigation for green spaces, industrial water, and ecosystem support for urban rivers and wetlands, while also acting as a buffer against shocks like drought, heavy rainfall, and pollution. The paper proposes a classification of cities by their hydrogeological settings (coastal, volcanic, alluvial, karst, cold-climate, arid) to better understand common issues: salinization and subsidence in coastal and delta cities, rising water tables damaging infrastructure, contamination from sewers and industrial spills, and the subsurface urban heat island effect. Equally, it identifies a suite of sustainable management practices that turn groundwater from a hidden liability into a resilience asset, including city-scale monitoring networks, 3D hydrogeological models, managed aquifer recharge, sustainable drainage systems, low-enthalpy geothermal energy use, and the formal registration of emergency wells for post-disaster water supply. The analysis shows that groundwater's contribution to urban resilience lies in redundancy (diversifying water sources), mitigation (reducing flood and heat risks), and preparedness (public awareness and real-time data), and that integrating groundwater management into urban planning is essential for cities to adapt to climate change and rapid urbanization. Ultimately, the paper argues that a groundwater-resilient city is one that makes the invisible visible—through continuous monitoring, transparent data sharing, and proactive governance—thereby turning the subsurface into an active, managed component of the urban water cycle rather than a forgotten crisis waiting to happen.This review examines the critical but often overlooked role of urban groundwater in sustaining city services and enhancing resilience, drawing on examples from over 70 cities worldwide. It highlights how groundwater—though hidden beneath pavement and infrastructure—provides essential services such as drinking water supply, irrigation for green spaces, industrial water, and ecosystem support for urban rivers and wetlands, while also acting as a buffer against shocks like drought, heavy rainfall, and pollution. The paper proposes a classification of cities by their hydrogeological settings (coastal, volcanic, alluvial, karst, cold-climate, arid) to better understand common issues: salinization and subsidence in coastal and delta cities, rising water tables damaging infrastructure, contamination from sewers and industrial spills, and the subsurface urban heat island effect. Equally, it identifies a suite of sustainable management practices that turn groundwater from a hidden liability into a resilience asset, including city-scale monitoring networks, 3D hydrogeological models, managed aquifer recharge, sustainable drainage systems, low-enthalpy geothermal energy use, and the formal registration of emergency wells for post-disaster water supply. The analysis shows that groundwater's contribution to urban resilience lies in redundancy (diversifying water sources), mitigation (reducing flood and heat risks), and preparedness (public awareness and real-time data), and that integrating groundwater management into urban planning is essential for cities to adapt to climate change and rapid urbanization. Ultimately, the paper argues that a groundwater-resilient city is one that makes the invisible visible—through continuous monitoring, transparent data sharing, and proactive governance—thereby turning the subsurface into an active, managed component of the urban water cycle rather than a forgotten crisis waiting to happen.

   b. V2 paragraph:
      i. This review talks about the important role of urban groundwater in supporting city services and helping cities become more resilient, using examples from over 70 cities around the world. Even though groundwater is mostly hidden underground, it is still very important because it supplies drinking water, supports irrigation and industries, and helps maintain rivers and wetlands. The study also explains how groundwater can protect cities from problems like droughts, floods, and pollution. However, different cities also face issues such as saltwater contamination, land sinking, rising water tables, and pollution caused by sewage and industrial waste. To deal with these problems, the paper suggests better groundwater management practices like monitoring systems, groundwater recharge, sustainable drainage, geothermal energy use, and emergency wells during disasters. Overall, the study emphasizes that cities need to include groundwater management in urban planning through proper monitoring, data sharing, and effective governance so they can better handle climate change and rapid urban growth.

6. Gozun & De Castro (2026): Their study in the Philippines highlights that reliability is the primary driver of household satisfaction. It isn't enough to have a tap; if water only flows for two hours a day, the household still operates in a state of "water poverty."
   a. Where is this from?

7. Bibri & Krogstie (2024): They argue that digital transformation is the "glue" of the sustainable city. They advocate for a Symbiotic Relationship where data informs governance, which in turn improves service delivery.
   a. Citation:
      i. Bibri, S. E., Krogstie, J., Kaboli, A., & Alahi, A. (2024). Smarter eco-cities and their leading-edge artificial intelligence of things solutions for environmental sustainability: A comprehensive systematic review. Environmental Science and Ecotechnology, 19, Article 100330. https://doi.org/10.1016/j.ese.2023.100330
   b. DOI link: https://doi.org/10.1016/j.ese.2023.100330
   c. Tentative Summary:
      i. This comprehensive systematic review investigates how the convergence of Artificial Intelligence and the Internet of Things (AIoT) is shaping “smarter eco-cities”—urban environments that integrate AI and AIoT technologies with sustainability strategies to enhance environmental performance. The study synthesizes evidence from 235 documents using

a unified framework that combines aggregative, configurative, and narrative methods. It identifies the foundational interplay between urban paradigms (smart cities, eco-cities, smart eco-cities), data-driven technologies, and environmental solutions. AI and AIoT applications across energy conservation, water management, waste handling, sustainable mobility, biodiversity protection, air quality monitoring, and climate adaptation are mapped, demonstrating benefits such as real-time resource optimization, predictive maintenance, reduced emissions, and more responsive governance. The paper also delineates a spectrum of challenges, including the environmental footprint of large-scale computing infrastructure, data privacy and cybersecurity risks, algorithmic bias, and the need for explainable AI to maintain public trust. It emphasizes that while AIoT can transform urban environmental services—making them more efficient, adaptive, and equitable—successful implementation demands robust regulatory frameworks, interdisciplinary collaboration, and strong policy support aligned with the Sustainable Development Goals. Ultimately, the review offers a roadmap for researchers and practitioners to leverage AI and AIoT not as a panacea but as a set of powerful tools that, when responsibly managed, can significantly improve the state of urban environmental services and advance the transition toward truly sustainable and intelligent cities.This comprehensive systematic review investigates how the convergence of Artificial Intelligence and the Internet of Things (AIoT) is shaping “smarter eco-cities”—urban environments that integrate AI and AIoT technologies with sustainability strategies to enhance environmental performance. The study synthesizes evidence from 235 documents using a unified framework that combines aggregative, configurative, and narrative methods. It identifies the foundational interplay between urban paradigms (smart cities, eco-cities, smart eco-cities), data-driven technologies, and environmental solutions. AI and AIoT applications across energy conservation, water management, waste handling, sustainable mobility, biodiversity protection, air quality monitoring, and climate adaptation are mapped, demonstrating benefits such as real-time resource optimization, predictive maintenance, reduced emissions, and more responsive governance. The paper also delineates a spectrum of challenges, including the environmental footprint of large-scale computing infrastructure, data privacy and cybersecurity risks, algorithmic bias, and the need for explainable AI to maintain public trust. It emphasizes that while AIoT can transform urban environmental services—making them more efficient, adaptive, and equitable—successful implementation demands robust regulatory frameworks, interdisciplinary collaboration, and strong policy support aligned with the Sustainable Development Goals. Ultimately, the review offers a roadmap for researchers and practitioners to leverage AI and AIoT not as a panacea but as a set of

powerful tools that, when responsibly managed, can significantly improve the state of urban environmental services and advance the transition toward truly sustainable and intelligent cities.

   d. V2 Paragraph:
      i. This study reviews how Artificial Intelligence (AI) and the Internet of Things (IoT) are helping create smarter and more sustainable eco-cities. By analyzing 235 studies, the paper explains how AI and AIoT technologies improve urban environmental services such as energy conservation, water management, waste disposal, transportation, air quality monitoring, and climate adaptation. These technologies help cities become more efficient through real-time monitoring, predictive maintenance, reduced emissions, and faster decision-making. However, the study also points out several challenges, including high energy use from computing systems, privacy and cybersecurity concerns, and possible bias in AI systems. The paper emphasizes that while AIoT has great potential to improve sustainability and urban services, its success depends on proper regulations, cooperation between different fields, and strong government support. Overall, the study concludes that AI and AIoT can become valuable tools in building smarter and more sustainable cities when they are used responsibly and effectively.

8. Gerlach et al. (2023): This research identifies the Enabling Environment. Technical engineering is easy; what is hard—and more important—are the legal and financial frameworks that allow a utility to be self-sustaining and autonomous from political cycles. Gerlach, C., Kühn, C. D., Mathiassen, A. B., Kristensen, C. L., & Starrfelt, R. (2023). *The face inversion effect or the face upright effect? Cognition, 232*, Article 105335. https://doi.org/10.1016/j.cognition.2022.105335
   a. Where is this from? From here? 10.1016/j.cognition.2022.105335
   b. Tentative paragraph:
      i. The study examined the "face inversion effect" (FIE), a phenomenon in which people have more difficulty recognizing faces when they are shown upside down compared to other objects. Researchers questioned whether this effect is truly unique to faces or if it is influenced by people's familiarity with recognizing upright faces in everyday life. To test this, the study compared how participants recognized inverted faces and objects using memory and object-recognition tasks that did not rely heavily on identifying individual differences within categories. The findings showed that inversion affected both faces and objects similarly, suggesting that the difficulty may stem more from familiarity with upright viewing conditions rather than from a face-specific cognitive process.

9. Jones, H., Cummings, C., & Nixon, H. (2014). Services in the city: Governance and political economy in urban service delivery (ODI Discussion Paper). Overseas Development Institute.

a. From:
https://odi.org/en/publications/services-in-the-city-governance-and-political-economy-in-urban-service-delivery/
b. Tentative paragraph:
    i. This discussion paper analyzes how governance and political economy factors shape urban service delivery in developing countries, focusing on solid waste management, water supply, transport, and urban health. Using frameworks of sector characteristics and common governance constraints, the authors show that while urban areas often enjoy greater resources and political attention than rural areas, these advantages do not automatically translate into equitable or high-quality services for all residents. The paper reveals how high population density, land scarcity, and social polarisation intensify externalities and create political market imperfections—such as clientelism, corruption, and populist subsidies—that skew provision toward visible, politically rewarding activities and away from less visible but essential functions like waste treatment or infrastructure maintenance. A diversity of public, private, and informal providers, especially in rapidly growing and informal settlements, leads to severe policy incoherence and monitoring challenges. The transient and politically marginalised nature of many urban poor populations further weakens collective action to demand better services. Across all four sectors, the study finds that effective service delivery hinges not only on technical capacity but on aligned political incentives, coherent institutional mandates, and mechanisms for accountability and coproduction. It concludes by identifying research gaps, particularly around under-studied services, the role of developmental political leadership, and the need for comparative, outcome-focused analyses to inform programming in informal settlements.
c. V2 Paragraph:
    i. This discussion paper examines how governance and political factors affect urban service delivery in developing countries, particularly in areas like waste management, water supply, transportation, and urban health. The study explains that although cities often receive more resources and government attention than rural areas, this does not always lead to fair or high-quality services for everyone. Problems such as corruption, clientelism, weak infrastructure maintenance, and politically motivated subsidies often influence how services are provided. The paper also points out that the presence of many public, private, and informal service providers can create confusion and weak coordination, especially in informal settlements. In addition, many urban poor communities struggle to demand better services because they are politically marginalized and highly mobile. Overall, the study concludes that improving urban services depends not only on technical solutions but also on strong governance,

clear institutional responsibilities, political commitment, and accountability between governments and communities.

10. Cheema, G. S. (2018). Access to urban services for inclusive development in Asia (Policy Brief). Swedish International Centre for Local Democracy.
    a. From: https://www.local2030.org/library/567/Access-to-Urban-Services-for-Inclusive-Development-in-Asia.pdf
    b. Tentative paragraph:
        i. This policy brief synthesizes findings from research in nine cities across five Asian countries—India, Indonesia, China, Vietnam, and Pakistan—on how democratic local governance shapes access to urban services for marginalized groups. It highlights that rapid urbanization has produced an “urban divide,” with the poor facing severe deficits in water, sanitation, housing, waste management, energy, transport, and health. Two overarching forms of exclusion are identified: institutional inequality, including unequal access to land, housing, and basic services, and structural inequality, encompassing barriers to participation for women, migrants, minorities, youth, and the elderly. The brief distils four policy priorities for inclusive service delivery: (1) strengthening local governments through democratisation, decentralisation, and grassroots engagement; (2) embedding transparency and accountability via participatory budgeting, anti-corruption bodies, procurement reform, and right-to-information legislation; (3) ensuring the active participation of migrants, women, and minorities in planning and decision-making, and tailoring policies to their specific circumstances; and (4) inclusive planning for peri-urban areas to address jurisdictional fragmentation, low institutional capacity, and service gaps. It underscores that equitable access cannot be achieved without collaborative approaches between citizens and municipal authorities, and that city-to-city learning can accelerate the replication of innovations and good practices.
    c. V2 Paragraph
        i. This policy brief looks at how local democratic governance affects access to urban services for marginalized groups across nine cities in India, Indonesia, China, Vietnam, and Pakistan. It explains that rapid urban growth has created a clear “urban divide,” where poorer communities often lack adequate access to basic services like water, sanitation, housing, transport, energy, waste management, and healthcare. The brief highlights two main types of inequality: institutional barriers, such as unequal access to land and services, and structural barriers that limit participation for groups like women, migrants, minorities, youth, and the elderly. To address these issues, it outlines key priorities, including strengthening local governments through decentralization and citizen participation, improving transparency and accountability through tools like participatory budgeting and anti-corruption measures, ensuring inclusive

decision-making for marginalized groups, and improving planning in peri-urban areas where governance is often fragmented. Overall, it stresses that fair access to urban services depends on strong cooperation between governments and citizens, along with sharing successful approaches between cities to improve outcomes for everyone.

11. Wang, H., & Ran, B. (2025). Localizing public service provision in megacities: Vertical and horizontal empowerment. Cities, 164, Article 106115. https://doi.org/10.1016/j.cities.2025.106115
    a. Tentative paragraph:
        i. This study investigates how Shanghai, as a megacity, piloted community new infrastructure projects across 17 sub-districts to localize public service delivery through a dual empowerment framework. Drawing on extensive interviews, ethnographic observation, and a resident survey, the authors reveal that vertical empowerment occurs through five mechanisms: aligning service goals across government levels, granting bounded autonomy to sub-district governments, differentiated resourcing tailored to local capacities, organized peer exchanges among implementing units, and periodic revision based on performance reviews. Horizontal empowerment is realized via coproduction, with residents moving from being merely informed or consulted (for government-funded basic services) to co-planning, co-designing, co-constructing, and co-assessing (for user-funded services). The study finds that neither vertical nor horizontal empowerment alone suffices; effective, equitable service localization hinges on their interplay, enhanced by intermediating non-profit and private actors who lubricate interactions between government and citizens. The case demonstrates that context-sensitive discretion, collaborative governance, and meaningful resident coproduction can reconcile city-wide standards with highly diverse local needs in the complex ecosystem of a megacity.
    b. V2 Paragraph:
        i. This study looks at how Shanghai tested community-level infrastructure projects across 17 sub-districts to make public services more local and responsive. It uses a “dual empowerment” framework to explain how service delivery was improved. Vertical empowerment comes from the government side, where higher levels of government set overall goals but also give sub-districts some controlled independence, adjust resources based on local needs, encourage learning between areas, and review performance regularly. Horizontal empowerment, on the other hand, involves residents being more directly involved in services, moving from just receiving information to actually helping with planning, design, building, and evaluation, especially in services they help fund. The study shows that neither top-down control nor community participation alone is enough. Instead, better results come when both work together, supported by NGOs and private actors that help connect governments and citizens.

Overall, the case shows that flexible governance and real resident involvement can help large cities like Shanghai balance standard city-wide systems with the different needs of local communities.

12. Stearns, F. (1980). Urbanization and public services. Georgia Journal of International and Comparative Law, 10(3), 495–526. https://digitalcommons.law.uga.edu/cgi/viewcontent.cgi?article=2004&context=gjicl
    a. Tentative paragraph:
        i. This paper examines the interconnected challenges of rapid urban population growth and the provision of public services—transportation, health, and education—in major cities of the developing world. It first traces the causes of urbanization, chiefly natural increase and rural migration, and identifies four distinct patterns of urban growth across Latin America, parts of Asia and North Africa, India and Indonesia, and sub-Saharan Africa. It then reviews government responses, including India's 1975 National Urbanization Policy Resolution, South Korea's comprehensive land-use and industrial decentralization laws, and Mexico's constitutional amendments and national urban development plan. Against this backdrop, the analysis turns to three service sectors. In transportation, it argues that rather than investing in costly subways or catering to private automobiles, developing cities should promote energy-efficient, low-cost modes—walking, cycling, traditional vehicles, and well-managed buses and minibuses—and manage demand through measures like Singapore's area licensing scheme. In health, it documents how poverty, malnutrition, and unsanitary conditions render the urban poor more vulnerable than rural populations and calls for a shift from hospital-based curative care to preventive, community-based services, improved nutrition, and expanded water and sewerage infrastructure. In education, it highlights how unequal facility distribution, high private costs, and a bias toward higher education exclude the poor, and recommends prioritizing low-cost basic literacy and vocational training, reducing subsidies to universities, and linking curricula to employment needs. The paper concludes that without massive, coordinated efforts that recognize the complementarity of rural and urban development, the already severe deficits in urban services will worsen dramatically as cities double in size.
    b. V2 Paragraph:
        i. This paper looks at the challenges that come with rapid urban population growth and how it affects public services like transport, health, and education in big cities in developing countries. It explains that urbanization is mainly driven by natural population growth and rural-to-urban migration, and that different regions experience this growth in different ways, such as in Latin America, Asia, North Africa, and sub-Saharan Africa. The paper also reviews how some governments have tried to manage these changes through policies on land use, decentralization, and urban planning. In transport, it suggests focusing

more on affordable and efficient options like walking, cycling, buses, and minibuses instead of expensive subway systems or private car use, along with demand control measures. In health, it points out that urban poor communities often face serious risks due to poor living conditions and recommends shifting toward preventive, community-based healthcare and better basic infrastructure like water and sanitation. In education, it highlights unequal access and high costs, suggesting a stronger focus on basic education and vocational training rather than prioritizing higher education. Overall, the paper argues that without coordinated efforts that connect rural and urban development, the gaps in urban services will continue to grow as cities keep expanding.

13. Furlong, K., Carré, M.-N., & Acevedo Guerrero, T. (2017). Urban service provision: Insights from pragmatism and ethics. Environment and Planning A: Economy and Space, *49*(11), 1–13. https://doi.org/10.1177/0308518X17734547
    a. Tentative Paragraph:
        i. This article argues that geographical debates on pragmatism and ethics offer a valuable yet largely overlooked lens for understanding urban service provision, particularly the complex agency of actors who negotiate tensions between higher-scale policies and on-the-ground realities. Drawing on Dewey, Varela, and others, it develops a framework where ethical action is situational, embodied, and shaped by long-term experience rather than rigid ideology. Through examples from Colombia, Nicaragua, South Africa, and elsewhere, the authors show how local officials, frontline workers, and community members engage in selective, normatively driven actions—such as tolerating informal connections, bypassing meters, or diverting waste—to mitigate the adverse effects of neoliberal reforms. A pragmatic perspective helps collapse the binary between structure and agency, revealing how power and ethical judgement are exercised in everyday “problematic situations” where formal rules confront entrenched inequalities and material constraints. The article ultimately underscores that while such agency can improve equity and service access, it remains bounded by wider processes of exclusion and is always contingent and context-specific.
    b. V2 Paragraph:
        i. This article argues that ideas from pragmatism and ethics can help us better understand how urban services are actually delivered in real life, especially when there’s tension between big policy rules and local conditions. It explains that ethical decisions are often not based on strict rules or ideology, but are shaped by experience, everyday realities, and what people think is right in specific situations. Using examples from places like Colombia, Nicaragua, and South Africa, the authors show how local officials, workers, and communities sometimes bend or adjust formal rules—like allowing informal water connections or working around strict waste systems—to make services work better for people. The paper

suggests that this kind of practical, on-the-ground decision-making helps reduce inequality and improve access in some cases. However, it also points out that these actions are still limited by larger systems of inequality and are very dependent on context, meaning they don't always lead to lasting or fully fair solutions.

14. Trejo Nieto, A. B., Niño Amezquita, J. L., & Vasquez, M. L. (2018). Metropolitan governance of public services in Latin America. Region, 5(3), 49–73. https://doi.org/10.18335/region.v5i3.224
    a. Tentative paragraph:
        i. This study compares metropolitan governance of public services—transport, solid waste collection, and water—in Bogotá, Lima, and Mexico City using a 3×3×3 framework (three cities, three services, three governance dimensions: coordination, financial sustainability, and coverage/quality). It finds that fragmentation undermines service equity and efficiency, especially in Mexico City, where all three services lack metropolitan coordination and suffer from financial precarity and uneven access. In contrast, Bogotá's services are "in consolidation" through historical annexation and cross-subsidy systems, while Lima's water is consolidated nationally, its waste sector shows nascent intermunicipal coordination, and its transport remains fragmented. The analysis reveals that no single metropolitan model guarantees success; instead, governance can adapt to local political, fiscal, and institutional conditions, and inter-jurisdictional collaboration can partially overcome administrative fragmentation. The paper underscores that effective metropolitan service delivery depends less on formal government structures than on flexible provision and production arrangements that can evolve across sectors.
    b. V2 Paragraph:
        i. This study compares how public services are governed in three major Latin American cities—Bogotá, Lima, and Mexico City—focusing on transport, solid waste collection, and water supply. It uses three key dimensions for comparison: coordination between different government levels, financial stability, and the overall coverage and quality of services. The findings show that when governance is fragmented, services tend to be less fair and less efficient, which is especially clear in Mexico City where coordination is weak and access is uneven across all sectors. Bogotá performs better overall, with services gradually improving through long-term integration and cross-subsidy systems, while Lima shows mixed results, with strong national water management but more fragmented transport and developing coordination in waste services. The study concludes that there is no single best model for metropolitan governance. Instead, effective service delivery depends on flexible cooperation between institutions that can adapt to local political and financial conditions, even when formal structures are not fully integrated.

Pia-Full Synthesized Content

Who Cities Work For and Who They Don't: The Urban Services Divide

Introduction

Urbanization has become one of the defining forces of the 21st century, reshaping how billions of people live, work, and access opportunity. Cities are often portrayed as engines of economic growth, technological innovation, and cultural development. Yet beneath the image of modern skylines and smart infrastructure lies a growing divide between communities that benefit from urban development and those that remain excluded from it. Across the world, millions of urban residents still lack reliable access to clean water, sanitation, transportation, electricity, housing, and digital connectivity. This imbalance, commonly referred to as the *Urban Services Divide*, reflects deeper inequalities embedded within infrastructure systems, governance structures, and patterns of spatial development.

Urban services are more than technical systems hidden beneath roads and buildings. They shape economic mobility, public health, environmental resilience, and overall quality of life. Access to reliable utilities determines whether individuals can pursue education, participate in the labor market, or recover from climate-related disasters. In many rapidly urbanizing cities, unequal service distribution reinforces cycles of poverty by forcing vulnerable populations to rely on informal, expensive, and unstable alternatives.

This project examines how infrastructure, governance, technology, and planning influence the distribution of urban services and opportunities. Drawing from academic research, multimedia sources, and case studies from both developed and developing cities, the discussion explores how cities can transition toward more inclusive, resilient, and sustainable systems through integrated planning, digital transformation, and community participation.

Blog Section: Critical Perspectives

1. Bridging the Gap: Resilient Infrastructure for All

Infrastructure is often described as the silent engine of the city, noticed only when it fails. At its core, infrastructure consists of the networks that allow cities to function, including roads, transport systems, water pipelines, drainage systems, and electrical grids. These systems create the foundation upon which economic activity and social life depend. However, within the context of the Urban Services Divide, infrastructure is not simply a collection of physical structures. It is a system that determines who gains access to opportunity and who remains excluded from it.

For residents living in underserved or informal communities, the absence of reliable infrastructure creates a constant struggle for survival. Without formal access to water, electricity, sanitation, or transportation, households spend significant amounts of time and money securing basic needs. This reduces opportunities for education, employment, and economic mobility. In many cities, the lack of infrastructure effectively creates invisible borders that separate economically connected populations from marginalized communities.

This inequality produces what economists describe as the “poverty premium,” where low-income residents pay more for lower-quality services. In cities such as Nairobi and Manila, informal settlers may pay several times more per liter of water purchased from private vendors than wealthier households connected to municipal systems. Climate change further intensifies these inequalities. Flooding, extreme heat, and infrastructure failures disproportionately affect vulnerable communities that lack resilient drainage systems, flood barriers, or disaster-resistant housing. In this context, resilient infrastructure becomes more than a development goal; it becomes a form of economic protection that prevents temporary crises from evolving into long-term poverty traps.

2. The Digital and Data Evolution of City Utilities

Modern cities are increasingly shaped by the relationship between infrastructure and data. Today, “bits” are becoming just as essential as “bricks” in the management of urban systems. Traditionally, utility management relied on reactive approaches where repairs occurred only after infrastructure failures became visible. However, advances in Artificial Intelligence (AI), the Internet of Things (IoT), and digital monitoring systems now allow cities to shift toward predictive management models.

Through sensors and real-time monitoring, utility providers can detect leaks, energy fluctuations, and infrastructure stress before major disruptions occur. This allows governments to reduce maintenance costs, improve efficiency, and strengthen long-term urban resilience. Digital transformation is especially important for rapidly growing cities with limited financial resources, where efficient management of existing infrastructure can be as valuable as constructing new systems.

Despite these advantages, digital transformation also introduces new forms of inequality. When smart technologies are concentrated only in wealthier districts, informal communities become invisible within urban data systems. Areas lacking digital monitoring are often excluded from future investment, planning, and service upgrades. In effect, the digital divide reinforces the existing urban services divide.

Inclusive digital governance therefore becomes essential. Data-driven systems must represent the realities of the entire city rather than only its formal sectors. When implemented equitably, digital infrastructure can reduce non-revenue water losses, improve energy reliability, strengthen disaster preparedness, and support modern service economies. Ultimately, technology should function as a tool for inclusion rather than another layer of urban exclusion.

3. Structural Transformations to Solve Urban Inequality

Urban inequality is often physically embedded within the structure of the city itself. Housing, transportation, employment opportunities, and public services are frequently distributed unevenly across urban space, forcing low-income populations to live far from economic centers. This spatial separation creates what scholars describe as a “spatial trap,” where workers spend excessive amounts of time and income commuting to jobs, reducing productivity and limiting upward mobility.

Traditional urban planning approaches frequently treat transportation, housing, sanitation, and infrastructure as isolated sectors. However, cities function as interconnected systems where failures in one area inevitably affect others. Integrated urban planning recognizes that improving public transportation, for example, also improves labor access, housing opportunities, and economic participation.

One example of this approach is the expansion of Bus Rapid Transit (BRT) systems into underserved areas. Reliable and affordable transportation does more than move people efficiently; it connects marginalized communities to formal labor markets and educational opportunities. Similarly, multi-purpose infrastructure projects such as parks that double as flood-control systems generate both environmental and social benefits. These integrated investments create higher long-term social returns by improving resilience, mobility, and public wellbeing simultaneously.

Ultimately, physical connectivity is closely tied to social mobility. A city that remains physically divided through unequal infrastructure and disconnected services will also remain economically divided.

Research Synthesis

Digital Transformation and Governance

The relationship between technology and governance is explored extensively in the work of Das (2024), which argues that smart cities require more than advanced technologies alone. Digital tools such as AI, IoT systems, and data platforms become effective only when paired with transparent and accountable governance structures. Using examples such as Singapore’s Smart Nation Initiative and Estonia’s X-Road platform, the research demonstrates how digital governance can improve efficiency, reduce corruption, and strengthen service delivery.

The study emphasizes that technology functions as a catalyst rather than a solution in itself. Without inclusive governance, digital systems risk reinforcing existing inequalities by prioritizing formally connected populations over underserved communities.

The Enabling Environment for Water

Trimmer et al. (2023) challenge the assumption that water access is purely a technical engineering issue. By studying six successful cities in the Global South, the researchers found that inclusive water provision depends largely on governance, financing, and institutional accountability. Successful cities combined efficient utility management with pro-poor policies such as subsidies and the removal of land tenure requirements for service connections.

The research highlights that sustainable and equitable water access requires an “enabling environment” where legal frameworks, political commitment, and financial systems support long-term service provision. Engineering solutions alone are insufficient without governance systems capable of maintaining and expanding access fairly.

Hybridization and Community Co-production

Moretto et al. (2023) examine how residents and governments often work together to fill gaps in formal infrastructure systems. In rapidly growing cities such as Dar es Salaam and Cochabamba, communities frequently develop localized solutions including shared water systems, storage networks, and informal service arrangements connected to municipal infrastructure.

This process of “hybridization” reflects the reality that urban service delivery is rarely fully centralized or fully informal. Instead, cities operate through overlapping systems involving governments, communities, and private actors. While these arrangements improve accessibility, the research also warns that modernization efforts can unintentionally exclude low-income communities if local systems are ignored or replaced entirely.

The findings suggest that resilient cities should strengthen and formalize collaborative systems rather than impose rigid one-size-fits-all models of infrastructure delivery.

The Role of Urban Groundwater

La Vigna (2022) highlights the often-overlooked importance of urban groundwater in supporting resilient cities. Groundwater functions as a hidden but critical resource that supports drinking water supplies, urban ecosystems, agriculture, and climate adaptation. It also serves as a natural buffer during droughts and extreme weather events.

However, urban groundwater systems face increasing threats from industrial pollution, overuse, rising water tables, and land subsidence. Poor groundwater management can damage infrastructure and intensify environmental risks. The study emphasizes the importance of integrating hydrogeological planning into urban development through monitoring systems, sustainable drainage, groundwater recharge, and real-time environmental data.

A resilient city must therefore manage not only visible infrastructure above ground but also the invisible environmental systems below it.

Synthesis: The Future of Our Cities

Addressing the Urban Services Divide requires moving beyond the binary distinction between “formal” and “informal” systems. The future of resilient urban development lies in technical and governance hybridization, where decentralized and community-led initiatives are integrated into formal planning systems rather than excluded from them.

Inclusive policies that prioritize underserved communities can reduce the long-term effects of spatial inequality and climate vulnerability. Investments in resilient infrastructure, equitable transportation, digital inclusion, and climate adaptation help prevent the formation of urban poverty traps while improving economic productivity and public wellbeing.

The long-term vision for sustainable urban development is reflected in the Compact City Model, where mixed-use, higher-density communities reduce the cost of service delivery while improving accessibility and environmental sustainability. In this model, cities are designed not only for efficiency and growth, but also for equity and resilience.

Blog Section: Remaining Perspectives

4. Circular Urban Metabolism: Turning Waste into Wealth

Traditional urban economies often follow a linear model in which resources are extracted, consumed, and discarded. As cities continue to grow, this approach becomes increasingly unsustainable. The concept of urban metabolism offers an alternative perspective by treating cities as interconnected systems where resources circulate continuously rather than ending as waste.

Through circular systems, wastewater can be converted into energy, organic waste can support urban agriculture, and recycled materials can be reintegrated into local economies. This shift toward circular urban metabolism is not only an environmental strategy but also an economic one. By reducing dependence on imported resources and generating localized industries, cities can create new forms of employment and improve long-term sustainability.

For underserved communities, decentralized waste-to-energy systems and localized recycling initiatives can also improve access to essential services. In this sense, circular infrastructure transforms public health risks into opportunities for resilience and economic inclusion.

5. Inclusive Governance: Managing the Informal City

Informal settlements are often treated as temporary problems or spaces outside formal urban planning. However, these communities frequently function as centers of economic activity, adaptation, and self-organization. Inclusive governance recognizes that governments alone cannot fully provide urban services in rapidly expanding cities. Instead, effective urban management requires collaboration between state institutions and local communities.

Through participatory planning and co-production models, governments can support community-led initiatives involving water access, drainage systems, transportation, and public safety. This approach shifts the role of government from being the sole provider of services to becoming an enabler of locally driven solutions.

Inclusive governance also depends on inclusive data systems. Informal communities are frequently excluded from official urban maps and datasets, making them invisible within planning processes. Participatory mapping and digital inclusion initiatives allow underserved populations to become visible within urban decision-making systems.

Ultimately, cities function most effectively when governance structures reflect the diversity and realities of the populations they serve.

Case Study Box: Manila’s Urban Services Divide

Manila: A City of Contrasts

Metro Manila reflects many of the challenges associated with the Urban Services Divide. As one of Southeast Asia’s largest megacities, it serves as a center of finance, education, and economic activity while simultaneously revealing the unequal distribution of urban services and opportunities.

In many informal settlements, households lack reliable access to piped water and instead depend on private water vendors who charge significantly higher prices than formal utility providers. This creates a “poverty premium” where low-income communities pay more for lower-quality services.

Transportation infrastructure also reflects deep inequality. Severe traffic congestion and fragmented transport systems increase commuting times for workers living far from economic centers, reducing productivity and limiting access to stable employment opportunities.

Climate vulnerability intensifies these problems further. Informal settlements are often located near waterways and flood-prone zones where drainage systems are weak or absent. During typhoons and periods of heavy rainfall, flooding disrupts transportation, contaminates water sources, damages homes, and increases health risks. In this way, environmental shocks frequently become economic shocks for already vulnerable households.

At the same time, Manila demonstrates the importance of community adaptation and hybrid service systems. Residents often organize localized drainage improvements, shared water systems, and informal transportation networks to compensate for gaps in formal infrastructure. These practices illustrate how resilience is frequently built through cooperation between communities and formal institutions.

Ultimately, Manila demonstrates that the Urban Services Divide is not simply a question of infrastructure shortages. It is fundamentally about who cities are designed to serve, who remains excluded from formal systems, and how governance can either reinforce or reduce inequality.

Conclusion

The Urban Services Divide reflects far more than shortages in infrastructure. It represents deeper inequalities in governance, planning, economic opportunity, and environmental resilience. Access to water, sanitation, transportation, energy, and digital systems determines who benefits from urban growth and who remains marginalized within rapidly expanding cities.

The research and case studies explored throughout this project demonstrate that resilient and inclusive urban development requires integrated solutions that combine technological innovation, participatory governance, climate adaptation, and equitable infrastructure investment. Whether through smart utilities, community-led systems, circular urban metabolism, or inclusive planning, the future of cities depends on their ability to serve all residents rather than only the formally connected population.

Ultimately, sustainable urban development is not simply about constructing smarter cities or taller skylines. It is about creating urban systems that provide dignity, opportunity, security, and resilience for every community. A city succeeds not when only its economic centers thrive, but when all residents are meaningfully connected to the services that allow urban life to flourish.

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